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Feeding Patterns in Grasshoppers (Orthoptera: Acrididae): Factors Influencing Diet Specialization

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Summary. Diets of grasshopper species from two arid grassland communities in Trans-Pecos, Texas, were determined by gut analysis. Species-specific food plant choice and niche breadths are presented for each of these species. As a group, grasshoppers range from monophagous to polyphagous feeders although most species fall in the oligophagous to polyphagous group. Phylogenetic constraints are evident such that gomphocerinae are primarily grass feeders while melanoplinae feed predominantly on forbs; the oedipodinae show less clearcut tendencies.

Feeding patterns are remarkably constant from site to site and overall, community niche breadth distributions between sites do not differ greatly. Individual species tend to eat the same plant species at various sites and maintain similar niche breadths. Species with relatively specialized diets tend to feed on predictable plant species such as grasses and long-lived perrenial forbs.

Grasshopper feeding patterns present some problems to the current theory of herbivore diet specialization since forb feeding melanoplines tend to be polyphagous (contrary to predictions). Life history patterns unrelated to tracking host plants may explain some aspects of diet breadth since diet selectivities are presumably adjusted according to the probability of finding suitable food plants.

Introduction

Herbivore feeding patterns (the number, specific taxa, and types of food plants) are of great interest because of their direct relevance to applied ecological problems and because herbivore feeding modes are not usually discussed in general foraging behavior models. The diversity of insect species and their feeding behaviors makes it difficult to formulate generalizations about the evolution

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of host specialization by herbivorous insects. Even so, advances are being made at several levels of biological organization (sensory to ecological) and a general theory of herbivore feeding is emerging (see reviews in Barton-Browne, 1975; Bernays and Chapman, 1974; Chapman, 1974; van Emden, 1973; Feeny, 1976; Gilbert, 1978; Otte and Joern, 1977; Rhoades and Cates, 1976).

Insect herbivore feeding patterns are influenced by physical, chemical, climatic, and biotic factors (see Brues, 1946, in addition to the above references). Much variation within and between insect groups (e.g., butterflies and grasshoppers) is the rule (Gangwere, 1961, 1972; Futuyma, 1976; Slansky, 1976; Uvarov, 1977) making it difficult to determine what evolutionary pressures produce a given feeding pattern (Otte and Joern, 1977). In particular, explaining the evolution of specialized versus generalized diets is still unresolved.

Combinations of approaches, primarily for lepidoptera, have yielded a robust, beginning theory of herbivore feeding (Feeny, 1976; Gilbert, 1978; Otte and Joern, 1977; Rhoades and Cates, 1976). Yet, some investigators (Otte, 1975; Futuyma, 1976; Fox and McCauley, 1972; Slansky and Feeny, 1977) have raised empirical objections to certain theoretical conclusions so it is appropriate to subject the theory to more empirical testing. Much of feeding theory is based on studies from lepidoptera which as a class are relatively specialized in their feeding. From a comparative standpoint then, the more polyphagous grasshoppers are ideal subjects.

Grasshoppers show much species-specific variation in food plant use (Anderson and Wright, 1952; Gangwere, 1961; Mulkern 1967; Mulkern et al., 1969; Otte and Joern, 1977; Sheldon and Rogers, 1978; Ueckert and Hansen, 1971; Uvarov, 1977) which ranges from absolute specialists to extreme generalists. Despite the wide ranges of plant taxa in their diets, grasshoppers are selective (Mulkern, 1967; Otte and Joern, 1977; Uvarov, 1977). For example, many potential food plants escape destruction under heavy infestation even when more palatable plants are scarce (Riley, 1875; Mulkern, 1967). Crop analysis studies (Bernays and Chapman, 1970; Gangwere, 1961; Mulkern et al., 1969; Otte and Joern, 1977; Ueckert and Hansen, 1971) as well as laboratory and field observations (Anderson and Wright, 1952; Clark, 1948; Gangwere, 1961) have documented these trends in selectivity.

Food plant choice is intricately entwined with many aspects of a species' biology and strongly affects several components of fitness such as fecundity, development, and survivorship (Mulkern, 1967; Uvarov, 1977). Preferences have also been shown to be correlated with several of the above components of fitness although certain anomalous trends are sometimes seen as in the relationship between alfalfa and *Melanoplus bivitattus* (Pfadt, 1949). Alfalfa is preferred even though feeding studies indicate grasshoppers raised exclusively on alfalfa have greatly reduced fecundity and survivorship. Such relationships indicate the non-trivial importance of food selection to the general biology of grasshopper species as well as some of the problems to be overcome in explaining feeding patterns. In this example, these species probably have not evolved together making the results suspect in assessing the evolution of food choice.

Food plant suitability is related to gain-cost relationships associated with each available plant species in an environment. Costs are more difficult to

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measure but usually represent nutrients and energy which could potentially be used for reproduction but aren't. As outlined in Otte and Joern (1977), plant choice by grasshoppers is undoubtedly influenced by secondary chemicals, nutrients (content, concentration, and distribution), sensory capabilities of insects and their use in finding hosts, competitors, environmental uncertainty, predation, historical consequences and phylogenetic constraints. Each of these factors can potentially shunt energy and nutrients away from reproduction or reduce the amount of energy (nutrient) which is gathered and thus reduce the amount available for allocation.

Chemical defense mechanisms coupled with the predictability with which a plant species may be found by an herbivore in time and space form the major framework of a theory of plant defense against herbivore pressure (Feeny, 1976; Rhoades and Cates, 1976). A plant's apparency (the ease with which herbivores may locate it in time and space) is strongly associated with the type of defense chemistry employed. Plants exhibiting degrees of apparency which are very predictable and long-lived or very abundant plants exhibit chemically similar defense systems (Feeny, 1976). In apparent plants, high molecular weight quantitative chemical systems are employed. An herbivore must expend much energy detoxifying such compounds (depending on the concentration) before the herbivore is able to assimilate the nutrients from the meal. The effect of these compounds is dosage-dependent and difficult to overcome evolutionarily. Oak tannis (Feeny, 1970) or phenolic resins found in desert creosote bushes (Rhoades and Cates, 1976) are examples of this tactic. Recent studies on Australian *Eucalyptus* trees which have high concentrations of phenolic compounds indicates some insect herbivores may readily deal with quantitative chemical defenses (Fox and McCauley, 1977; Morrow, 1977).

Unapparent plants, which are unpredictable in time or space, employ other chemical methods (Feeny, 1977; Rhoades and Cates, 1977). A large component of the plant's defensive tactic depends on not being found by herbivores. Qualitative chemical defenses are employed by unapparent plant species. Chemicals which are inexpensive to produce (low molecular weight) but highly variable between plant species are included in this category. Such chemical species tend to be more acutely toxic than the quantitative system but are more easily overcome in an evolutionary sense by herbivores since they act in an all-or-none fashion (Feeny, 1976; Rhoades and Cates, 1976). Thus, once an herbivore species evolved physiological mechanisms to counteract a qualitative defense, it may feed with impunity on such a plant species.

Comparative feeding studies from geographically distinct areas permit one to examine the influence of the above factors on the evolution of acridid feeding patterns. In this paper, feeding behaviors of grasshoppers from Trans-Pecos, Texas arid grasslands are compared with other studies from northern U.S. grasslands and southwestern deserts. Effects of gross ecological factors and circumstantial arguments concerning plant nutrient and chemical attributes allow a consideration of general hypotheses concerning insect herbivore feeding.

Questions relating to grasshopper feeding behavior revolve around specific feeding patterns, proximate cues influencing plant choice, and evolutionary relationships explaining grasshopper feeding selectivities. Important questions in-

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clude: Are grasshoppers typically generalists or specialists? Are populations with broad feeding niches made up of individuals with wide preference ranges or specialist individuals which differ? What proximate mechanisms influence food selection? Are phylogenetic effects prevalent? How do grasshopper assemblages use the available plant resources? Do generalist grasshoppers eat different types of plants than specialists? What factors might account for such a difference? Are grasshoppers with narrow food niches more likely to be obligate or facultative specialists? Do forb feeding species show different feeding patterns than grass-feeding species? than mixed feeding species? Do geographic and intercommunity feeding patterns exist? Here I summarize grasshopper feeding patterns which relate to these questions.

Study Sites

Collections were made from May until November in 1974 and 1975 near Alpine, Texas, a mountainous area possessing a mosaic of vegetation types. Although located at the northern edge of the Chihuahuan Desert, study areas are dominated by arid grassland plant species.

Altuda is an approximately 30–40 hectare ungrazed site surrounded by sheep and cattle grazed ranchland 24 km east of Alpine. This site has not been grazed for at least five years and probably much longer. Since the site is near a railroad right of way, the possibility of disturbance by fire is great. Vegetation is composed of many plant species (ca. 60 at peak density) dominated by blue gramma (*Bouteloua gracilis*). The level substrate consists of dark colored, calcareous soils with a surface soil of greyish brown calcareous loam.

Marathon is adjacent to a railroad right of way 16 km east of Altuda and 13 km west of the town of Marathon. Most sampling was done on a hectare plot situated in a larger relatively undisturbed pasture approxiately 40 hectares in area. This site was also not grazed in the recent past although a portion of the plot was burned sometime during the spring, 1975. The grama grasses *Bouteloua hirsuta* and *B. curtipendula* are dominant. Substrate consists of dark colored calcareous soil with greyish brown loam at the surface.

Altuda and Marathon sites are typical of the arid grassland plains of this area. Characterized by low cloud cover and unpredictable, summer rains, temperatures typically exceed 40° C during the summer period. Average annual precipitation is about 36 cm with about 80% falling between May and October. August and September are the wettest months (U.S. Climatological Summary). More climatological information is found in Joern (1977).

The seasonal appearance of grasshopper species varies in both species composition and diversity. Peak species densities are found during the end of August and first portion of September. Phenological patterns include early-occurring and late-occurring species, grasshoppers found only during a short period in August, and species found throughout the duration of the study (Joern, 1977). Seasonal changes in subfamily composition are also seen. Gomphocerines dominated the community composition during the entire study in terms of proportion of species. Gomphocerines and oedipodines reach peak species abundances dur-

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ing July and August while the melanoplines reach peak diversity during late August through November with only a few species present during May through July.

Methods

Field observations and gut analyses provide information on food plant utilization by these grasshoppers. Adult grasshoppers were killed immediately after collection in ethyl acetate charged vials and the foreguts removed and placed in 70% ethanol shortly after death (an hour or less). Plant fragments from the gut were compared with permanently mounted fragments taken from plant voucher specimens collected in the area (Isley and Alexander, 1949; Mulkern and Anderson, 1959). Trichomes (small hairs and spines) and cell wall structure (including stomata patterns) were used to identify the plants. Some plant species combinations were indistinguishable and are lumped in the analysis.

Relative abundances of the plant material in the diet were estimated. The most abundant material in the gut was scored and all other plant species present in the gut recorded. These data are weighted to yield a resource utilization spectrum for food use by the grasshopper community. In the present analysis, abundant plant species in the gut are weighted twice as much as the other plants. This scheme is followed for two reasons: 1) The most abundant plant is almost immediately apparent. In fact, the most abundant plant easily dominates the plant species composition of a particular gut sample. 2) When other weighting schemes are employed (such as weighting all plant species in the gut equally or weighting the most abundant plant three times the importance of the other plants in the gut before calculating the relative abundances of certain plant taxa in the diet, resulting measures differed little. Percentages of plant species in the community diet calculated from weighting schemes of $3X-1X$ differed less than 5% from the percentage plant species calculated using a weighting scheme considering only abundant plants and excluding others. The weighting scheme chosen for the present analysis is used because it appears to be conservative for comparing diet breadths of monophagous or polyphagous groups.

Relative availability of plants in the environment is determined by estimating plant cover using the point intercept method (Grieg-Smith, 1964). Cover estimates for each plant species are summed and converted to species-specific relative abundances. Sample points within a one hectare area study plot were determined from a random numbers table and four measurements were taken at each point in a cruciform pattern one meter from the randomly chosen center. Wiens (1974) uses a similar sampling program in examining habitat requirements of coexisting grassland birds.

Feeding specialization (niche breadth) is measured using the number of plant taxa eaten or a niche breadth index which weights the amount of each resource used. An information-theoretic niche breadth measure (B) is given as:

$$B = \exp(H')$$

where

$$H' = -\sum_i p_i \ln p_i .$$

In these relationships, p_i represents the proportion of each plant species in the diet. Standardized niche breadths may be calculated by dividing B by the number of available resources.

This index is a population measure where individual choices over a short time interval are lumped. Because of this, care must be taken in interpreting the results since while some populations may be composed of specialist individuals eating different plant taxa, the population measure would indicate a wide species-specific niche breadth. This would not be a true reflection of the feeding biology. A second problem of estimating preference is noted in Otte (1975). If an insect eats large quantities of one plant and only small quantities of another, it is sometimes erroneously assumed that the former is preferred. This is not true if only small amounts of the later plant and large amounts of the former are required for a complete diet. Since grasshoppers often develop

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successfully on rarely eaten plants (Otte, 1975), relative abundances of different food plants in the diet probably reflect preferences coupled with availabilities of plants in the environment.

For comparative purposes, data presented in Mulkern et al. (1969) and Ueckert and Hansen (1971) are evaluated using this niche breadth index. This facillitates comparison among geographic localities, especially when standardized niche breadths are used.

Results

Diet Specialization

1) Grasshoppers exhibit varying degrees of plant selectivity (Table 1). Some species usually feed on only one or at most a few plant species (e.g., *Tropidolophus formosus* or *Mermiria bivittata*) while other species feed on a wide variety of plant species (e.g., *Trimerotropis pallidipennis* or *Dactylotum variegatus*). In the selection process, a large number of plant families are sampled representing varied growth forms, structural attributes, and chemical compositions. Comparison between Altuda and Marathon indicates grasshoppers include a similar array of plant species in the collective community diets.

The degree of selectivity is measured by comparing the proportion of various plant species found in the diet and the abundances of available plant taxa (Table 1). Abundances of plant taxa not eaten are not included. No species exhibits a lack of selectivity.

Some grasshoppers shift the relative proportions of plant species in their diet at different sites. *Opeia obscura* feeds almost exclusively on *Bouteloua gracilis* at Marathon but includes relatively large proportions of *Buchloe dactyloides* and *Leptoloma cognata* (at the expense of *B. gracilis*) at Altuda. *Aulocara femoratum, Syrbula fuscovittata, Arphia crassa,* and *Leprus cyaneus* also exhibit this trend between sites. Other grasshopper species consume very similar diets at both sites (e.g., *Ageneotettix deorum* and *Eritettix variabilis*) while still other species (e.g., *Melanoplus desultorius*) completely change their diets. At Marathon, *M. desultorius* eats much *Xanthocephalum texanum* whereas *Croton pottsii* dominates its diet at Altuda and *X. texanum* is excluded. These plant species are found in similar abundances at the two sites.

2) Degree of specialization is most easily evaluated either by counting the number of food plants used or by constructing an index which compiles the number of plant taxa used and weights their relative abundance in the diet (information-theoretic index). Results for two sites are presented in Table 2. Information-theoretic measures range from 1.0 to 13.8 at Marathon and from 1.0 to 14.6 at Altuda. Although similar numbers of plant taxa are used at the two sites, niche breadths are slightly lower at Altuda. However, a comparison of niche breadths between the sites using a nonparametric Mann Whitney U test is not significant ($P=0.16$, two-tailed).

3) Grasshopper populations are composed of individuals foraging in a similar manner rather than being composed of individuals feeding selectively on different plant taxa. The mean number of plant species per individual gut in relation to the niche breadth index demonstrates this relationship (Fig. 1). A high Spearman rank correlation is found when food niche breadth is compared with

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the average number of species per gut ($r_s=0.85$ at Marathon and $r_s=0.73$ at Altuda; $p<0.01$).

4) Niche breadths for species common to both study sites are strikingly constant. Grasshoppers tend to show similar niche breadths at the two sites (Fig. 2); a Spearman rank correlation for niche breadth of 12 species from both sites is significant ($r_s=0.84$, $p<0.01$). Actual niche breadth values are very similar as well except for *Arphia crassa* which changes from 4.81 at Altuda to 1.88 at Marathon. *A crassa* eats twice as many plant species at Altuda and shows a much lower affinity for *Bouteloua gracilis* at Altuda even though this plant is relatively more common than at Marathon. However, if *A. crassa* does not discriminate between *B. gracilis* and *B. hirsuta, Bouteloua* is more abundant at Marathon and may explain the strong reliance on these species and the associated low niche breadth at Marathon.

Most other grasshopper species common to both sites have similar diets although additional plant taxa may be included. As mentioned earlier, *Opeia obscura, Aulocara femoratum, Leprus cyaneus,* and *Syrbula fuscovittata* choose slightly different diets at the two sites. Typical forb-feeding species exhibit very similar niche breadths between the two sites. However, although the average niche breadths for this group are very similar between sites, species tend to include a few more species in their diet at Marathon. Species-specific comparisons of standardized niche breadths among these Trans-Pecos, Texas, sites with other geographically separated grassland sites (Mulkern et al., 1969; Ueckert and Hansen, 1971) exhibit remarkable consistency (Table 3) even though the resource bases among the sites are quite different.

Some of these grasshopper feeding patterns are particularly noteworthy. *Opeia obscura* has a narrow standardized niche breadth over the five sites for which records are available. A similar trend is seen for *Hesperotettix viridis.* Some species such as *Ageneotettix deorum, Campylacantha olivacea, Melanoplus lakinus,* and *Phoetaliotes nebrascensis* exhibit changes in the food plants taken when different sites are examined. Since these species are either grass or forb feeding, plant group affinity (suggesting either phylogenetic or historical influences) doesn't seem to explain the result.

5) A great diversity of niche breadths exists among grasshopper species ranging from specialists to those which are generally oligophagous or polyphagous. Figure 3 presents the food niche breadth distribution for a number of grasshopper species from several geographic localities (all grassland). Desert grasshoppers,

Table 1 (pp. 332–335). Diets of grasshopper species from Altuda and Marathon. Use of particular plant species is presented as the proportion the plant species is used by the grasshopper relative to all other plant species in the grasshopper's diet. Only plant species found in the collective diets at each site are presented. Relative abundances represent the frequency of a particular plant species in the environment relative to all other plant species present at the site. Plant families are coded: 1. *Poaceae;* 2. *Acanthaceae;* 3. *Amaranthaceae;* 4. *Asclepiadaceae;* 5. *Asteraceae;* 6. *Boraginaceae;* 7. *Brassicaceae;* 8. *Choenopodiaceae;* 9. *Convolvulaceae;* 10. *Euphorbiaceae;* 11. *Fabaceae;* 12. *Linaceae;* 13. *Loganaceae;* 14. *Malvaceae;* 15. *Nyctaginaceae;* 16. *Onagraceae;* 17. *Polemoniaceae;* 18. *Polyganaceae;* 19. *Rhamnaceae;* 20. *Scrophulariaceae;* 21. *Verbenaceae.* Grasshopper subfamilies are ordered as gomphocerinae, odeipodinae and melanoplinae. Diet resource states noted by letters are found in the diets but can not be matched to a known plant species

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Table 1. Altuda

Family		1	1	1	1	1	1	1	1	1	1	1	1	2	5	5	5	5	5	5	5	5	6	9	10	11	11	12
Food Plant Species	Sample Size	*Aristida hamulosa*	*Bouteloua curtipendula*	*Bouteloua gracilis*	*Bouteloua hirsuta*	*Buchloe dactyloides*	*Eragrostis intermedia*	*Leptaloma cognata*	*Muhlenbergia sp.*	*Panicum obtusum*	*Scleropogon brevifolius*	*Setaria leucophila*	*Sporobolus sp.*	*Dychoriste linearis*	*Conyza coulteri*	*Eupatorium greggii*	*Iva dealbata*	*Machaeranthera scabrella*	*Ratibida columnaris*	*Xanthocephalum texanum*	*Xanthocephalum sp.*	*Zinnia grandiflora*	*Cryptantha mexicana*	*Convolvulus equitans*	*Croton pottsii*	*Hoffmanseggia glauca*	*Prosopis glandulosa*	*Linum rigidum*
Relative Abundance		*	*	0.22	*	0.13	0.05	0.03	0.10	*	0.06	0.01	*	*	*	*	*	*	*	*	*	*	*	*	*	*	0.04	*
Grasshopper Species																												
Ageneotettix deorum	21			0.75		0.05		0.11																	0.02			
Aulocara femoratum	29			0.28	0.04	0.37		0.11			0.18																	
Boopedon nubilum	38			0.55	0.03	0.18		0.09			0.04	0.08																
Cordillacris crenulata	44			0.74	0.06	0.05		0.02																	0.02			0.04
Eritettix variabilis	33			0.77		0.07			0.01								0.01											0.05
Opeia obscura	44	0.02		0.66		0.19		0.10	0.02																0.01			
Philbostroma quadrimaculatum	21			0.21		0.48		0.05	0.05		0.19																	
Psoloessa texana	38			0.73		0.08		0.01			0.01		0.07															
Syrbula fuscovittata	12			0.67		0.25					0.08																	
Arphia crassa	20			0.31	0.05	0.14		0.33																				
Leprus cyaneus	19			0.28		0.30	0.04	0.19			0.13														0.04			
Trachyrachys kiowa	52			0.65		0.17		0.03			0.13	0.02																
Trimerotropis pallidipennis	101			0.20		0.04		0.07									0.08					0.10		0.12	0.04			
Tropidolophus formosus	16																								1.0			
Campylacantha olivacea	14																0.33			0.06	0.06							
Dactylotum variegatum	76			0.01									0.01		0.02		0.16	0.07	0.02	0.01				0.08	0.05	0.01		
Melanoplus arizonae	20					0.04											0.04			0.04				0.10	0.15			
Melanoplus desultorius	25			0.03											0.03				0.12			0.22		0.02	0.15			
Melanoplus gladstoni	43			0.02									0.03		0.01	0.01			0.02	0.02				0.20	0.18		0.01	
Melanoplus glaucipes	17			0.06										0.03					0.03	0.03					0.41	0.06		
Melanoplus lakinus	29		0.03	0.09			0.09										0.04	0.03					0.03	0.05	0.15			
Phoetaliotes nebrascensis	29			0.13		0.05	0.03		0.02	0.02							0.03								0.03			0.06

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Table 1. Altuda (continued)

Family	16	16	16	17	18	19	19	21																							
Food Plant Species	*Calyophus hartwegii*	*Gaura coccinea*	*Oenothera sp.*	*Gilia rigida*	*Polygala longa*	*Condalia ericoides*	*Ziziphus obtusifolius*	*Verbena wrightii*	*Unidentified voucher No. 28*	Unknown Forbs A	B	C	D	E	F	G	H	J	M	N	O	P	Q	R	S	Insect Parts	Undeterminable grass	Undeterminable forb
Relative Abundance	*	*	*	*	*	*	*	*	*																						
Grasshopper Species																															
Ageneotettix deorum																0.07															
Aulocara femoratum																															0.02
Boopedon nubilum																															
Cordillacris crenulata															0.01																0.01
Eritettix variabilis							0.01																					0.05			0.03
Opeia obscura																															
Philbostroma quadrimaculatum																															
Psoloessa texana										0.003																		0.02	0.01		0.03
Syrbula fuscovittata																							0.02								
Arphia crassa						0.10																	0.07								
Leprus cyaneus																													0.02		
Trachyrachys kiowa														0.01															0.01		
Trimerotropis pallidipennis		0.02											0.02	0.03						0.11	0.01			0.02	0.01		0.02	0.01	0.04	
Tropidolophus formosus																															
Campylacantha olivacea			0.03				0.06				0.09	0.03	0.18			0.15															
Dactylotum variegatum	0.01	0.08		0.01		0.02	0.01					0.06	0.06	0.01	0.04		0.05				0.01	0.01							0.02	0.15
Melanoplus arizonae									0.08		0.08	0.27																0.02	0.06	0.04
Melanoplus desultorius				0.03	0.05				0.03			0.05	0.07		0.02		0.02		0.05									0.02		0.07
Melanoplus gladstoni									0.16		0.01	0.19		0.02		0.03														0.05
Melanoplus glaucipes	0.03					0.06		0.09	0.06				0.09																0.03	0.02
Melanoplus lakinus									0.17		0.04	0.16				0.03		0.02												0.07
Phoetaliotes nebrascensis												0.13	0.40															0.05		0.05

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Table 1. Marathon

Family		1	1	1	1	1	1	1	1	1	1	1	1	2	3	4	5	5	5	5	5	5	6	7	7	8	9	10
Food Plant Species	*Sample Size*	*Aristida wrightii*	*Bouteloua curtipendula*	*Bouteloua gracilis*	*Bouteloua hirsuta*	*Buchloe dactyloides*	*Ergarostis intermedia*	*Hilaria mutica*	*Leptoloma cognata*	*Muhlenbergia sp.*	*Scleropogon brevifolius*	*Setaria leucophila*	*Sporobolus sp.*	*Unidentified species*	*Tridestromia lanuginosa*	*Asclepius sp.*	*Dyssodia acerosa*	*Gaillardia pinnatifida*	*Machaeranthera scabrella*	*Iva ambrosiaefolia*	*Xanthocephalum texanum*	*Zinnia grandiflora*	*Cryptantha mexicana*	*Lepidium lasiocarpum*	*Lesquerella fendleri*	*Choenopodium sp.*	*Convolvulus equitans*	*Argythamnia neomexicana*
Relative Abundance		0.03	0.19	0.10	0.22	0.01	0.16	*	*	*	*	*	*	*	*	*	*	*	*	*	0.01	*	*	*	*	*	*	*
Grasshopper Species																												
Acrolophitus variegatus	8								0.05															0.11	0.63			
Ageneotettix deorum	38		0.03	0.66	0.08						0.04		0.08															
Aulocara femoratum	32			0.56	0.28								0.03															
Cibolacris parviceps	17														0.23			0.13					0.05	0.15		0.08		
Eritettix variabilis	24			0.73		0.18				0.09																		
Heliaula rufa	67	0.05		0.67	0.02	0.19																						
Mermiria bivittata	33			0.93		0.02	0.04																					
Opeia obscura	23			0.94		0.04																						
Parapomala wyomingensis	12			0.88	0.06																							
Pedioscirtetes maculipennis	7															1.0												
Psoloessa texana	87			0.65	0.09	0.08	0.01			0.04																		
Syrbula fuscovittata	17			0.94	0.06																							
Arphia aberrans	8			0.43		0.05																						
Arphia crassa	15			0.80	0.13			0.07																				
Hadrotettix trifasciatus	42			0.03														0.02			0.03		0.04		0.08	0.02		0.06
Leprus cyaneus	22		0.04	0.56	0.20	0.09					0.04																	
Trachyrachys aspera	6			0.81	0.19																							
Trimerotropis pallidipennis	28			0.08	0.01		0.03	0.01				0.03	0.01	0.04						0.05	0.07		0.03		0.01			0.01
Xanthippus corallipes	11			0.70	0.10															0.07								
Hesperotettix viridis	40																0.04				0.70	0.05						
Melanoplus arizonae	18				0.07														0.05		0.05							
Melanoplus desultorius	57				0.01									0.06	0.01			0.08			0.20		0.01			0.01		
Melanoplus gladstoni	19			0.02																0.09			0.09		0.08			
Phoetaliotes nebrascensis	14					0.07	0.14														0.03							

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Table 1. Marathon (continued)

Family	10	11	11	11	13	14	15	16	16	17	18	20	21																		
Food Plant Species	*Croton pottsii*	*Acacia constricta*	*Cassia roemeriana*	*Dalea greggii*	*Buddleja scordioides*	*Sida filicaulis*	*Allionia incarnata*	*Calyophus hartwegii*	*Guara coccinea*	*Gilia rigidula*	*Polygala longa*	*Castilleja sp.*	*Verbena wrightii*	Unidentified voucher No. 28	Unknown voucher No. 1	Unknown Forbs												Insect Parts	Undeterminable
																B	C	D	F	G	H	I	J	K	L	M	N		
Relative Abundance	*	*	*	*	*	*	*	*	*	*	*	*	*	*	*														
Grasshopper Species																													
Acrolophitus variegatus	0.11																			0.11									
Ageneotettix deorum																				0.04								0.03	0.03
Aulocara femoratum																	0.05	0.05											0.03
Cibolacris parviceps		0.03								0.03						0.03	0.05						0.05	0.05		0.03			0.10
Eritettix variabilis																													
Heliaula rufa	0.02																			0.01									0.01
Mermiria bivittata																													0.02
Opeia obscura																	0.03												0.03
Parapomala wyomingensis																													
Pedioscirtetes maculipennis																	0.06	0.01		0.01			0.02		0.01			0.01	
Psoloessa texana																													
Syrbula fuscovittata																													
Arphia aberrans	0.19																0.29												0.05
Arphia crassa																													
Hadrotettix trifasciatus	0.15					0.02		0.01		0.01	0.11		0.01			0.10	0.02						0.22			0.02		0.03	0.03
Leprus cyaneus																										0.02			0.04
Trachyrachys aspera																													
Trimerotropis pallidipennis	0.07									0.03	0.03						0.16	0.08				0.03	0.08	0.03			0.03	0.08
Xanthippus corallipes	0.10										0.03																		
Hesperotettix viridis				0.03			0.04				0.03										0.03							0.01	0.07
Melanoplus arizonae	0.27						0.02							0.05		0.05	0.12	0.05	0.05				0.10	0.02				0.02	0.07
Melanoplus desultorius	0.02		0.01	0.01	0.02		0.01		0.02		0.09	0.01	0.01	0.01		0.01	0.02						0.03	0.02	0.03		0.15		0.16
Melanoplus gladstoni	0.04						0.03				0.13					0.02	0.11							0.04				0.09	0.30
Phoetaliotes nebrascensis	0.03														0.38		0.03												0.03

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Table 2. Food niche breadths. An information-theoretic index is used to calculate a weighted measure of food plant use by the grasshopper species at both Altuda and Marathon. Resources used is an unweighted niche breadth measure. H' and B are defined in the Methods. Standardized niche breadths (B/N) are computed where N represents the number of plant species eaten at a particular site

	Sample	Resources used	H'	B	B/N
Altuda					
Ageneotettix deorum	21	5	1.18	3.25	0.06
Aulocara femoratum	29	5	1.41	4.11	0.07
Boopedon nubilum	38	6	1.30	3.67	0.07
Cordillacris crenulata	44	7	0.92	2.50	0.04
Eritettix variabilis	33	7	0.82	2.27	0.04
Opeia obscura	44	5	1.02	2.77	0.05
Philbostroma quadrimaculatum	21	6	1.40	4.04	0.07
Psoloessa texana	38	7	0.86	2.36	0.04
Syrbula fuscovittata	12	3	0.82	2.26	0.04
Arphia crassa	20	6	1.57	4.81	0.09
Leprus cyaneus	19	6	1.57	4.79	0.09
Trachyrachus kiowa	52	5	1.00	2.71	0.05
Trimerotropis pallidipennis	101	16	2.47	11.81	0.21
Tropidolophus formosus	16	1	0.00	1.00	0.02
Campylacantha olivacea	14	9	1.90	6.70	0.12
Dactylotum bicolor	76	22	2.68	14.64	0.26
Melanoplus arizonae	20	10	2.26	9.62	0.17
Melanoplus desultorius	25	16	2.42	11.26	0.20
Melanoplus gladstoni	43	15	2.08	8.03	0.14
Melanoplus glaucipes	17	11	1.94	6.98	0.12
Melanoplus lakinus	29	13	2.29	9.83	0.18
Phoetaliotes nebrascensis	29	11	1.88	6.59	0.12
MARATHON					
Acrolophitus variegatus	8	5	1.17	3.21	0.06
Ageneotettix deorum	38	7	1.15	3.17	0.06
Aulocara femoratum	32	5	1.19	3.29	0.06
Cibolacris parviceps	17	12	2.34	10.38	0.19
Eritettix variabilis	24	3	0.76	2.13	0.04
Heliaula rufa	67	6	0.93	2.55	0.04
Mermiria bivittata	33	3	0.27	1.31	0.02
Opeia obscura	23	2	0.17	1.19	0.02
Parapomala wyomingensis	12	3	0.37	1.44	0.03
Pedioscirtetes maculipennis	7	1	0.00	1.0	0.02
Psoloessa texana	87	11	1.31	3.70	0.07
Syrbula fusovittata	17	2	0.23	1.25	0.02
Arphia aberrans	8	4	1.20	3.31	0.06
Arphia crassa	15	3	0.63	1.88	0.03
Hadrotettix trifasciatus	42	18	2.46	11.76	0.21
Leprus cyaneus	22	6	1.21	3.36	0.06
Trachyrachys aspera	6	2	0.49	1.63	0.03
Trimerotropis pallidipennis	28	21	2.62	13.76	0.25
Xanthippus corallipes	11	5	1.00	2.72	0.05
Hesperotettix viridis	40	8	1.02	2.78	0.05
Melanoplus arizonae	18	13	2.17	8.76	0.16
Melanoplus desultorius	57	23	2.55	12.76	0.23
Melanoplus gladstoni	19	10	2.16	8.64	0.16
Phoetaliotes nebrascensis	14	10	1.79	5.97	0.11

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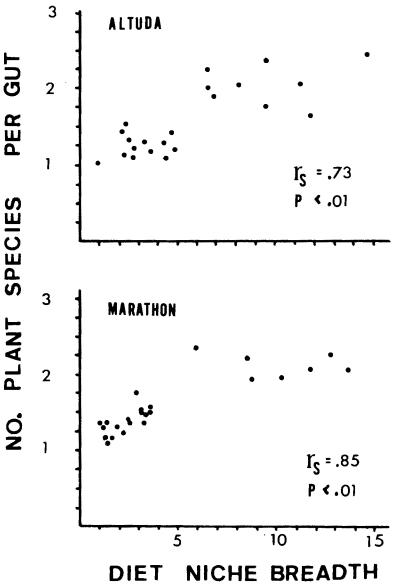


Fig. 1. Relationship between food niche breadth and number of plant species per gut. An information theoretic index (*B*) estimates niche breadth. Each point represents the mean number of plants per gut for different grasshopper species in relation to the species niche breadth at that site

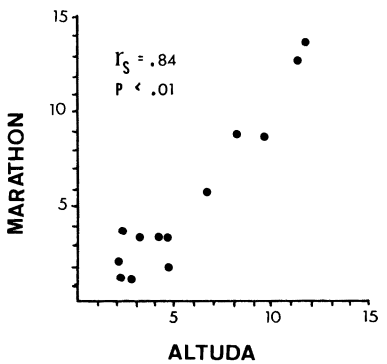


Fig. 2. Comparisons of food niche breadth for species at different localities. A positive Spearman Rank Correlation for niche breadths of species at Altuda and Marathon exists

by contrast, have a much lower feeding niche breadth (Otte and Joern, 1977) and a higher percentage of monophagous species than do comparable grassland sites.

6) Phylogenetic constraints on feeding behavior are evident. Gomphocerines tend to be strictly grass feeders while melanoplines feed on forbs; oedipodines are often mixed feeders although grasses usually predominate. In addition to feeding restrictions associated with plant taxa, ranges of plant species taken are also different among the subfamilies (Table 4). Gomphocerines have the lowest niche breadth and melanoplines the largest. This trend is consistent at both study sites.

7) Diet is related to grasshopper life histories, especially the length of time the species is present during the growing season. For analysis, species are cate-

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Table 3. Food niche breadth comparisons between communities. Standardized food niche breadths (information − theoretic = B/N) are presented for as many sites as possible. Species included are those with information from at least two sites

	Altuda	Marathon	Kansas[a]	North Dakota[a]	N. Platte[a]	Colorado[b]
Ageneotettix deorum	0.06	0.06	0.06	0.13	0.07	0.12
Cibolacris parviceps	–	0.19	–	–	0.04	–
Aulocara femoratum	0.07	0.06	–	–	–	–
Eritettix variabilis	0.04	0.04	–	–	–	–
Opeia obscura	0.05	0.02	0.04	0.03	0.03	–
Philbostroma quadrimaculatum	0.07	–	–	–	0.03	–
Psoloessa texana	0.04	0.07	–	–	–	–
Syrbula fuscovittata	0.04	0.02	–	–	–	–
Arphia crassa	0.09	0.03	–	–	–	–
Leprus cyaneus	0.09	0.06	–	–	–	–
Trachyrachys aspera	–	0.03	–	–	–	–
Trachyrachys kiowa	0.05	–	–	–	0.03	0.03
Trimerotropis pallidipennis	0.21	0.25	–	–	–	–
Xanthippus corallipes	–	0.05	–	–	–	0.13
Hesperotettix viridis	–	0.05	0.08	0.09	0.05	–
Campylacantha olivacea	0.12	–	0.02	–	–	–
Melanoplus arizonae	0.17	0.16	–	–	–	–
Melanoplus desultorius	0.20	0.23	–	–	–	–
Melanoplus gladstoni	0.14	0.16	–	0.16	–	–
Melanoplus lakinus	0.18	–	–	–	0.02	–
Phoetaliotes nebrascensis	0.12	0.11	0.07	0.16	0.08	–

[a] Data from Mulkern et al. (1969)
[b] Data from Ueckert and Hansen (1971)

gorized as occurring early, late, all season, or only during a middle portion of the season. Diet niche breadths (Table 5) are lowest for those species found during the middle portion of the flight season only and highest for those grasshoppers found throughout the season. Species with flight periods early or late in the season have intermediate average niche breadths. However, only comparisons between all-season grasshoppers and either early and late groups are statistically significant. It is probable that mid-season occurring species are also different from species found throughout the year, but this is swamped by a small sample size.

Community Use Patterns

1) Community organization patterns based on food may be examined by comparing feeding patterns among separate localities. Food niche breadth distributions are compared between communities in Table 6. Other feeding studies are included for comparison (Mulkern et al., 1969; Ueckert and Hansen, 1971; Hansen

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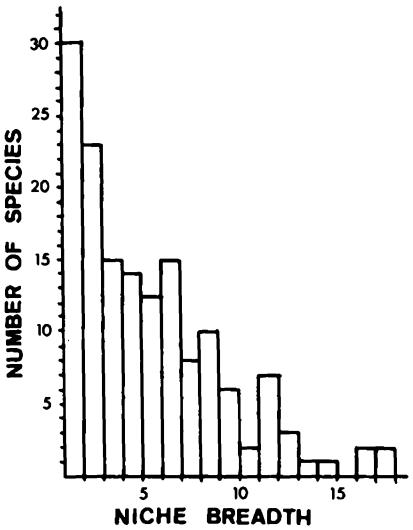


Fig. 3. Grasshopper species food niche breadths. The distribution of niche breadths from West Texas arid grasslands (this study), northern U.S. grasslands (Mulkern et al., 1969) and Colorado sandhills grasslands (Ueckert and Hansen, 1970) are lumped. An information-theoretic niche breadth index is used. Whole integer niche breadth values are included in the next category (e.g., 1 lies between 1 and 2)

Table 4. Mean food niche breadth by locality and subfamily. Standard deviations are given in parentheses

Subfamily	Number of species	Mean resources used	Mean H'	Mean B
Altuda	22	8.73 (5.06)	1.54 (0.68)	5.73 (3.72)
Gomphocerinae	9	5.67 (1.32)	1.08 (0.25)	3.03 (0.76)
Oedipodinae	5	6.80 (5.54)	1.32 (0.91)	5.02 (4.11)
Melanoplinae	8	13.38 (4.24)	2.18 (0.028)	9.20 (2.77)
Marathon	24	7.63 (6.11)	1.21 (0.81)	4.66 (4.01)
Gomphocerinae	12	5.0 (3.52)	0.82 (0.67)	2.89 (2.55)
Oedipodinae	7	8.43 (7.72)	1.36 (0.83)	5.49 (5.04)
Melanoplinae	5	12.80 (5.97)	1.94 (0.58)	7.78 (3.70)

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Table 5. Food niche breadth and grasshopper species phenology. Food niche breadths (B) are presented according to the flight period of the grasshopper species. Each class is compared with a Mann Whitney U statistic

	Niche breadths		
	Altuda	Marathon	Lumped
Early	2.39 $n=2$	4.22 $n=5$	3.69 $n=7$
Middle	5.12 $n=2$	1.22 $n=2$	3.17 $n=4$
Late	5.67 $n=14$	4.87 $n=14$	5.27 $n=28$
All Season	7.90 $n=4$	6.25 $n=3$	7.40 $n=7$

	Significance tests based on lumped data-probabilities			
	Early	Middle	Late	All Season
Early	–			
Middle	0.324	–		
Late	0.125	0.45	–	
All Season	0.037	0.158	<0.001	–

and Ueckert, 1970). Of the 21 pairwise comparisons among communites, 13 (62%) are statistically different at the 5% level (Mann Whitney U; two-tailed). Altuda is the most similar to all of the communities while Kansas, North Dakota, and Colorado sites are the least similar (if only grasshopper studies are compared – Hansen and Ueckert (1970) include other herbivores).

2) Plant abundance and utilization by the grasshopper community are only tenuously related. A positive correlation between plant species abundance (relative frequency) and the intensity of use by all grasshoppers in a community sometimes exists. At Altuda, a significant position correlation between community use and relative availability of plant species exists ($r=0.69$, $p<0.01$), a relationship which is not found at Marathon ($r=0.18$, $p>0.05$). A positive relationship between community resource use and resource availability is found for grasshoppers in the shrub-steppe region of Washington (Sheldon and Rogers, 1978).

3) Analysis of community resource utilization indicates only a small proportion of the available plant species are used in any appreciable amount (Table 7). The following comparisons are made using only those plants actually found in the collective community gut sample. The proportion of total available plant species which cumulatively account for 50% of community resource utilization and the proportion of species used at least 1% of the time are compared among communities. Only about 6.5% of the plant species account for over

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Table 6. Community food niche breadth comparisons. Standardized niche breadths based on information-theoretic indices are compared for several grasshopper communities. A table of probabilities associated with values of z in the normal distributions determined by a Mann-Whitney U statistic are presented for the niche breadth comparisons among these communities. Probabilities are based on two-tailed tests

	Marathon	Altuda	Kansas	North Dakota	North Platte	Colorado I	Colorado II
Marathon	–						
Altuda	0.19	–					
Kansas[a]	0.37	0.01	–				
North Dakota[a]	0.0003	0.23	0.0002	–			
North Platte, Nebr.[a]	0.62	0.14	0.27	0.03	–		
Colorado I[b]	0.02	0.27	0.004	0.77	0.03	–	
Colorado II[c]	<0.0001	<0.0001	<0.0001	<0.0001	0.008	<0.0001	–
Number different	3/6	2/6	4/6	4/6	3/6	4/6	6/6

[a] Data from Mulkern et al. (1969)
[b] Data from Ueckert and Hansen (1971)
[c] Data from Hansen and Ueckert (1970)

Table 7. Plant resource utilization by a grasshopper community. Plant species making up diets are ranked according to the contribution made to total grasshopper community utilization. The number of plants contributing at least 1% of use and the cumulative number of plants necessary until 50% of the diet is determined are compared among sites

Study Site	Total No. of plant resources used	No. of species until 50% of total Use obtained	% of total Spp. number	Cumulative % of overall use of these Spp.	Plant species greater than 1%	% of total plant Spp.
Altuda	56	3	5.4	52.8	15	26.8
Marathon	55	2	3.6	51.3	14	25.5
Mulkern[a] Kansas	37	2	5.4	73.0	12	32.4
Mulkern[a] N. Dakota	69	6	8.6	54.1	21	30.4
Mulkern N. Platte, NE.	68	6	8.8	52.3	22	32.4
Ueckert[b] and Hansen Colorado	47	3	6.4	58.6	17	36.2
Mean			6.4			30.6

[a] Data from Mulkern et al. (1969)
[b] Data from Ueckert and Hansen (1971)

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50% of the total community consumption while only 30% of the plants are used at least 1% of the time. Thus, about two-thirds of the plants eaten by grasshoppers are used only infrequently and in some cases, a very small number of plant species (Kansas) account for an extremely high proportion of total plant consumption.

Discussion

Grasshopper feeding patterns differ from other well-studied herbivore groups such as butterflies. A wide range of feeding niche breadths and switching among food types is sometimes seen when different communities are compared. Differences in the specific taxa and range of plants eaten by different grasshopper species may be influenced by community level competitive phenomena, phylogenetic constraints, abilities of grasshoppers to detoxify different plant defensive chemicals, grasshopper life history characteristics which are not related to the appearance of specific food plants, mobility and dispersal abilities of different grasshopper species, nutrient properties of potential food plants, or physiological attributes specific to individual grasshopper species. Determining the role each of these factors plays is difficult.

Phylogenetic Influences

Broad similarities among members of the same subfamily and striking differences between subfamilies suggest a strong phylogenetic component to feeding adaptations. Speciation events probably took place after a major switch to different types of food plants by the antecedent species. Although it is possible to break away from such a pattern, as seen in grass-feeding melanoplines or forb-feeding gomphocerines, such adaptive shifts are not the rule and most gomphocerines eat grasses and most melanoplines eat forbs. Similar results are presented by Otte and Joern (1977). The role of the mixed feeding oedipodines will prove interesting since species in this group often eat both grasses and forbs.

The presence of basically dichotomous patterns may represent the adaptive peaks suggested by Wright (1956). Morphological, physiological, or even behavioral adaptations may act as constraints making major switches between types of host plants very difficult. In assessing causality relationships of cryptic, specialist lepidoptera larvae and their food plants, Brower (1958) indicates the difficulty in ascribing either direction or importance of either factor in the evolution of diet specialization. Multi-factorial pressures could constrain further diversification within a group. Switches among plants within the same major group may present no unique problems, however, and costs associated with such a switch may be small compared to major changes. More information on the role of sensory systems and metabolic pathways associated with digesting and detoxifying chemicals from different plant taxa is needed before these possible costs may be assessed.

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Effects of Plant Chemistry

Defensive secondary chemicals and nutrient concentrations (especially nitrogen) undoubtedly affect host plant choice. These plant chemical attributes are related to plant species abundance and predictability in considerations of herbivore feeding theories (Feeny, 1976; Rhoades and Cates, 1976). Predictable species such as shrubs and trees presumably contain quantitative chemical defenses while unpredictable plant species usually contain qualitative defenses.

Melanopline feeding patterns are anomalous in the context of present theory. Members of this subfamily exhibit the greatest niche breadths, contrary to what is expected in a forb feeding group. If the food plants being consumed rely on qualitative defenses and each species is composed of different chemical arrays, present theory predicts melanopline species should specialize on specific food plants and evolve detoxification mechanisms to deal with the specific plant chemistry. Possibly, melanoplines circumvent the defense systems by eating only small portion of many plants thus keeping the toxic levels for each secondary compound below some critical level (Freeland and Janzen, 1974). To do this, they would have to eat many species, each of which had different secondary compounds. Compensatory mechanisms where low nitrogen concentrations result in an increased rate of food intake indicates this is a possible mechanism (McGinnis and Kasting, 1966; Slansky and Feeny, 1977). Slansky and Feeny (1977) showed reduced assimilation efficiency (dry weight) associated with increased intake in *Pieris rapae*.

Alternatively, forbs found in arid grassland habitats may represent predictable, late successional forms which rely on quantitative defenses (Feeny, 1976; Gilbert, 1978). If this were true, the wide range of feeding behaviors in melanoplines would not be unexpected since each plant species would be using similar chemical defense systems and grasshopper species would be equipped to handle defensive systems of many plant species easily since they would be so similar. Otte (1975) found such feeding patterns in five *Schistocerca* species where late successional plants were more palatable than early successional ones. These trends have been interpreted (Gilbert, 1978) in relation to qualitative versus quantitative chemical defense mechanisms; the late successional plant species are more acceptable because they employ the same series of quantitative plant compounds while early successional species incorporate unique qualitative defense chemistries against which the *Schistocerca* are not equipped to handle. However, quantitative chemicals are presumed to be dosage dependent and very difficult to overcome evolutionarily. Further study on the chemical defenses employed by these arid land plant species is needed before melanopline feeding behaviors may be explained. Present theory based on defensive chemical attributes alone may not be sufficient since it seems unlikely these forbs employ primarily quantitative defenses.

Grasses fall into the predictable plant category. Although little information is known abut the chemical properties of most grasses, especially mature plants, the often discussed structural attributes and high silica content in grasses may be an alternative tactic to chemical defenses. In addition, C_3 and C_4 photosynthetic pathways have structural correlates which may affect herbivore feeding

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efficiency. Highly vascularized bundle sheath cells surrounding available plant nutrients in plants employing C_4 pathways may make these species nutritionally less desireable (Caswell et al., 1973; Caswell and Reed, 1976). In the present study, however, this is not a factor affecting choice among grasses since all grasses employ C_4 photosynthetic pathways (Fry et al., 1978). Some seedling grasses do employ defensive chemicals which deter feeding (Bernays et al., 1974) and chemical arrays are known from various grasses (Hegnauer, 1963). Gomphocerines which do eat forbs are specialists on these plants as predicted (e.g., *Pedioscirtetes maculipennis*). Similarly, the forb feeding oedipodine *Tropidolophus formosus* is a specialist on *Croton pottsii*. Why the other grass feeding gomphocerine and oedipodine grasshoppers do not sample from all of the available grasses indiscriminantly is not known but may relate to either unknown chemical properties or nutrient deficiencies.

Host Plant Predictability

Grasshoppers may not be tracking food resources and life history phenologies or reproductive patterns (voltinism) may be driven by other agents. Feeding patterns may be affected nevertheless. If a species is found during the season when food plants are productive and abundant, specializing on a single food plant might be a safe and efficient feeding tactic. However, if a species is long-lived and experiences a large part of the season during a portion of the year when food choice is poor and many plants are senescing, feeding choices in terms of which plants are eaten or how many are eaten should be different. Food resource predictability could be very important to these long-lived species.

Multivoltine or long-lived species generally have broad feeding niches. Species such as *Trimerotropis pallidipennis, Hadrotettix trifasciatus, Dactylotum variegatum,* and *Melanoplus desultorius* are among the longest lived species and all have large niche breadths. *Hesperotettix viridis* is also found throughout the study period but feeds primarily on *Xanthocephalum texanum*. This plant is a perrenial, predictable species (verging on being woody) found throughout the season and contains a high concentration of aromatic chemical compounds which are undoubtably toxic. Long-lived gomphocerines include *Psoloessa texana* and *Opeia obscura*. Although niche breadths for these species are lower than those cited above, *P. texana* has one of the larger niche breadths for grass feeding species while *O. obscura* varies in resource use depending on the presence of the grass *B. gracilis*. At Marathon, *O. obscura* has a larger niche breadth and *B. gracilis* is less abundant than at Altuda where it uses only two grasses and feeds primarily on *B. gracilis*.

Early occurring grasshoppers have low niche breadths and high average resource use overlaps (Joern, 1977) while later-occurring species tend to have higher niche breaths. The predictability of grass may explain the relationship. Relatively high niche breadths for those species found in the latter portion of the season suggests these species do not track plant species and must be able to feed on a wide range of host plants. This period follows the summer rains and feeding patterns could reflect the high diversity of plant species which

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suddenly become available (although in a spatially uncertain manner since rainfall is very patchy in this area). Competitive interactions may also be invoked to explain the community feeding patterns since this midportion of the season contains the greatest number of grasshopper species which may coexist by using different plant species. Overlaps in specific resource utilizations are lower during this period than during other ones (Joern, 1977).

Responses to plant predictability, then, may take several forms. If an herbivore species is long-lived compared to most of the available plant species, it should eat a wide range of taxa during the course of its life span or specialize on a plant species which is highly predictable in space and time. Short-lived species may specialize on predictable plant species which occur at the same time or eat many species depending on the availability. For short-lived species to specialize on particular plant taxa, however, it must be able to track the species in time and space by responding to environmental cues associated with plant production or specialize on a long-lived plant species which is present during the grasshopper's life span.

Competitive Influences on Food Selection

Competition among species could result in host shifts by species coexisting in a community where food (or some attribute of food plants related to feeding or nutrition) is limiting. Ueckert and Hansen (1971) claim competition for food is the primary determinant of grasshopper diet organization. Sale (1975) has questioned this interpretation since these species tend to converge on a certain few resources rather than diverge. Since many species use the same plant species (ca. 7% of the plant species account for 50% or more of the community resource use), it appears segregation onto other plant species should take place if competition is an important factor.

Food plant utilization in Trans-Pecos grasshopper assemblages is hyper-dispersed (Joern and Lawlor, submitted), an expected result if competitive interactions are important (Schoener, 1974). Yet, such a pattern need not implicate competitive interactions. Secondary compounds could prevent a number of grasshopper species from eating particular plant species. When resource utilizations are reorganized randomly in simulations except that zeros in the resource use matrix are maintained (i.e., the grasshopper is not allowed to eat plants it normally does not) the actual hyper-dispersed distribution of feeding does not differ greatly from this reorganized version (Joern and Lawlor, submitted). Thus, such simulations indicate that a hyper-dispersed resource utilization does not necessarily imply competitive interactions. Since total resource utilization by the grasshopper community does not necessarily correspond to the availability of these resources, it is very likely competition does not proscribe the patterns of cummunity use for food. If competition is a strong influence, more variable species-specific feeding patterns and switching among available host plants at different sites would be expected, especially since the grasshopper taxonomic milieus which determine competitive interactions vary among sites. In addition, grasshopper species should use those food plants which are not exploited. Spe-

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cific food utilization patterns, then, probably reflect coevolutionary interactions among plant taxa and grasshoppers with competitive interactions among grasshopper taxa having minimal influence. However, dynamic interactions causing specific food plant choice are still unknown, although current theory (Feeny, 1976; Rhoades and Cates, 1976) has provided a framework of evolutionary tactics which will play an important role. For grasshopper diet selection though, factors other than those found in butterflies are at work and the chemical explanations will need to be modified.

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